

MCA102 Data Structures using C

Course Code	MCA102		
Course Name	Data Structures using C		
Course Type	Core/Lab		
Credit	4		
Semester	1		
Course Objectives	To introduce the concept of linear and nonlinear data structures. To implement the concepts using arrays and linked list To apply it to advanced data structures		
Pre-requisites	Bridge Course		
Course Details	Lecture Hours	Tutorial Hours	Practical Hours
	2		4

Course Outcomes (CO)

CO#	CO Description	Learning Domain*
CO1	To differentiate the linear and nonlinear data structures	U
CO2	Implement the various kinds of sorting and searching techniques.	C, A
CO3	To implement the concept of nonlinear data structures using arrays and linked list.	C, A
CO4	Familiarize the concept of advanced data structures like red black trees, AVL trees etc.	U
CO5	Implement the concept of balancing a tree and the rotations to do it.	C, A

* Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

CO-PSO-PO Mapping Matrix

CO- PO- PSO	PO1	PO2	PO3	PO 4	PO5	PO 6	PO7	PO 8	PO 9	PO10	PO11	PO12	PSO1	PSO 2	PSO 3
CO#1	3						1				2	2		2	
CO#2			3				1				2	2		2	
CO#3	3		2				1				2	2		2	
CO# 4			3				1				2	2		2	

CO#5			2				1				2	2		2	
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3- High correlation, 2- medium correlation, 1-low correlation, 0- no correlation

Course Content

Module No		Module Content	Hours Required
1	1.1	Introduction: Data Structures, Data Types, Structure.	6
	1.2	Arrays: Polynomial Representations, Polynomial addition, Polynomial Multiplication and sparse matrices	
	1.3	Stack: Definition and concepts, Operations on stacks.	
	1.4	Application of stacks- Infix to postfix conversion, Evaluation of Arithmetic Expression.	
2	2.1	Queue: Representation of queue, circular queue and double ended queue. Priority queue: implementation by array using Heap Sort	6
	2.2	Dynamic Memory Allocation Functions: malloc, calloc, realloc and free	
	2.3	Linked List: Operations – insertion, searching, removing, updating, sorting and reversing	
	2.4	Polynomial: Representations, Addition, Multiplication using Linked List	
3	3.1	Linear Data Structures: Linked stacks, Linked queues, Circular Linked List and Double Ended Queue, Doubly Linked List and Circular doubly linked list.	6
	3.2	Non-Linear Data Structures: Trees, Graphs.	
	3.3	Graph: Representation of Graph on Computer: Adjacency matrix and adjacency list, merits and demerits of graph representation	
	3.4	Searching: Linear Search, Binary Search	
4	4.1	Trees: Basic terminology, binary trees, binary search tree	6
	4.2	Binary search tree: Insertion, Deletion, searching and Traversal – inorder, pre-order and post-order.	
	4.3	Threaded Binary Tree: Operations	
	4.4	Balanced Trees: AVL Tree: properties, insertion, deletion and rotations	
5	5.1	Advanced Data Structures: Red black tree: properties.	6
	5.2	B-Trees: Data Structure on secondary storage, Definition of B trees, Basic operations on B Trees – searching, creating an empty node, splitting a node in B Tree, Inserting a key in to B Tree and Deleting a Key from a B Tree	
	5.3	Definition and Structure: B+ Trees	
	5.3	Data Structure for Disjoint Sets: Disjoint set operation, linked list representation of disjoint sets, Disjoint-set forests	

References (Text books/ Researches/ Online URLs)

Fundamentals of Data Structures in C by Horowitz, Sahni and Anderson-Freed.
Data Structures Through C in Depth by S.K Srivastava, Deepali Srivastava
Data Structures Using C Aaron M. Tenenbaum
Data Structures Using C, Reema Thareja

Instruments to evaluate the CO

Evaluated CO	Tool	Date of Evaluation
MCA102.1	WA, CAE1	
MCA102.2	WA, CAE1	
MCA102.3	CAE2	
MCA102.4	CAE2	
MCA102.5	Project (EC)	

Rubrics of Evaluation

CATEG ORY	Assess ed Course objecti ve	Assessment Rubrics (out of 5)					
		5	4	3	2	1	0
CAE1 and CAE2	1,2,3,4	Correctly answer the conceptual questions which are not straight forward, may be combining different concepts of data structures	Answer direct questions perfectly	Attempts direct questions and answers partially.	Concepts are not clear	Answered, but not correct	Didn't Attempt
Case Study/ Project / Assign ment	1,2,3	Perfect Solution with expected functionality	80% Completed Solution	50% correct Solution	Below 30% functionality	Below 10% functionality	0%

Detailed Session Plan

Note:

- (1) Specify the span of the session as 1 hour/ 2 hour
- (2) For a 3-credit paper, session plan for 45 hours and for a 4-credit paper, a session plan for 60 hours is to be given.

Session#	Topic	Model of Delivery/ Experiential Component	Related CO	Actual Date of the Session
1	Introduction: Data Structures, Data Types, Structure.	Lecture	MCA102.1	
2	Stack: Definition and concepts, Operations on stacks.	Hand – on session	MCA102.1	
3	Application of stacks- Infix to postfix conversion, Evaluation of Arithmetic Expression	Lecture	MCA102.1	
4	Queue: Representation of queue,	Hand – on session	MCA102.3	
5	circular queue and double ended queue.	Hand – on session	MCA102.3	
6	Priority queue: implementation by array using Heap Sort	Hand – on session	MCA102.3	
7	Arrays: Polynomial Representations, Polynomial addition, Polynomial Multiplication and sparse matrices	Lecture	MCA102.3	
8	Polynomial Multiplication and sparse matrices	Lecture	MCA102.3, MCA102.2	
Session#	Topic	Model of Delivery/ Experiential Component	Related CO	Actual Date of the Session
9	Dynamic Memory Allocation Functions: malloc, calloc, realloc and free	Hand – on session	MCA102.1	
10	Linked List: Operations – insertion, searching, removing, updating, sorting and reversing.	Hand – on session	MCA102.1	
11	updating, sorting and reversing.	Hand – on session	MCA102.3	
12	Polynomial: Representations, Addition, Multiplication using Linked List.	Hand – on session	MCA102.4	
13	CAET	Exam	MCA102.4	
14	Linear Data Structures: Linked stacks, Linked queues,	Hand – on session	MCA102.1, MCA102.3	

15	Circular Linked List and Double Ended Queue,	Hand – on session	MCA102.1, MCA102.3	
16	Doubly Linked List and Circular doubly linked list.	Hand – on session	MCA102.1, MCA102.3	
17	Non-Linear Data Structures: Trees, Graphs. Graph: Representation of Graph on Computer: Adjacency matrix and adjacency list, merits and demerits of graph representation	Hand – on session	MCA102.1, MCA102.3	
18	Searching: Linear Search, Binary Search	Hand – on session	MCA102.2	
19	Trees: Basic terminology, binary trees, binary search tree	Hand – on session	MCA102.1, MCA102.4	
Session#	Topic	Model of Delivery/ Experiential Component	Related CO	Actual Date of the Session
20	Binary search tree: Insertion, Deletion, searching and Traversal – inorder, pre-order and post-order.	Hand – on session	MCA102.4	
21	Threaded Binary Tree: Operations	Hand – on session	MCA102.4	
22	Balanced Trees: AVL Tree: properties, insertion, deletion and rotations	Hand – on session	MCA102.5	
23	CAE2	Exam		
24	Advanced Data Structures: Red black tree: properties.	Lecture	MCA102.5	
25	B-Trees: Data Structure on secondary storage, Definition of B trees, Basic operations on B Trees – searching	Lecture	MCA102.5	
26	creating an empty node, splitting a node in B Tree, inserting a key in to B Tree and Deleting a Key from a B Tree	Lecture	MCA102.5	
27	Definition and Structure: B+ Trees	Lecture	MCA102.5	
28	Data Structure for Disjoint Sets: Disjoint set operation,	Lecture	MCA102.5	
29	linked list representation of disjoint sets, Disjoint-set forests	Lecture	MCA102.5	
30	Revision	Lecture		

DATA STRUCTURES USING C -LAB CYCLE

1. Write programs to demonstrate the use of storage classes (local variable, global variable, static variable, register variable) in C.

2. Use a menu-driven program to insert, search, delete and sort elements in an array using functions (use global variables)
3. Use a menu-driven program to insert, search, delete and sort elements in an array using functions (use only local variables)
4. Search for all the occurrences of an element in an integer array (positions)
5. Sort the array elements in ascending order (minimum three functions: read, disp and sort)
6. Display the array elements in the same order using a recursive function
7. Display array elements in reverse order using a recursive function.
8. Write a program to Perform the addition of two matrix and Subtraction of one matrix from another
9. Write a program to perform multiplication of two matrix
10. Write a program to find the transpose of a matrix
11. Write a program to find the Determinant of a matrix (2x2 and 3x3)

Section 2: Stack

12. Implement stack operations using arrays

Session 3 - String

13. Read a String and Just print it in the reverse order
14. Read a String and Reverse the string in the same array itself
15. Read n Strings and display them in the ascending order.

Session 4 – Stack- Continued

16. Reverse a string using Stack
17. Convert an expression from infix expression to postfix using stack
18. Convert an expression from infix expression to prefix using stack
19. Evaluate an infix expression using stack
20. Evaluate an expression using stack by converting it into postfix before evaluating
21. A letter means pushes and an asterisk means pop in the following sequence. Give the sequence of values returned by the pop operations when this sequence of operations is performed on an initially empty LIFO stack.

E A S * Y * Q U E * * * S T * * * I O * N * * *

Session 5 - Sparse Matrix

- 22. Read and display a sparse matrix
- 23. Write a program to add two sparse matrix
- 24. Write a program to multiply two sparse matrix

Session - 6 Polynomial using array

- 25. Read a polynomial and display
- 26. Add two polynomials
- 27. Subtract two polynomials
- 28. Multiply two polynomials

Session-7

- 29. Demonstrate queue using array
- 30. Demonstrate circular queue using array.

Session 8

- 31. Implement heapsort
- 32. Demonstrate a priority queue using an array.

Session 9: Structure

- 33. Define a structure for dates with dd/mm/yyyy. Provide functions for reading, displaying and comparing two dates are equal or not
- 34. Define a structure for employees with eno, ename, esal and dno. Read n employees information and provide functions for the following:
 - a. Searching an employee by no
 - b. Sorting the employees by
 - i. Name
 - ii. Salary
 - c. Deleting an employee

Session 10: Polynomials using Structure

- 35. Read a polynomial and display it; use structure array

- 36. Add two polynomials
- 37. Subtract two polynomials
- 38. Multiply two polynomials

Session 11: Dynamic Memory Allocation

- 39. Implement a) malloc , b) calloc and c) free functions
- 40. Use malloc to read n integers and find the mean.
- 41. Use calloc to read n numbers and find the mode.
- 42. Declare a structure for Books having author_name and book_name. Create an array of books using a pointer variable. Provide functions for reading n books and displaying the same using pointers.
- 43. Use realloc to implement varchar for any length.
- 44. Demonstrate a linked list creation and display
- 45. Write a program with functions to insert a new node
 - a. at the beginning of a Singly Linked List.
 - b. At the end of the linked list
 - c. after a specified element in a linked list.
- 46. Write a program with functions to delete a node
 - a. From the beginning of the linked list
 - b. From the end of the linked list
 - c. The node with specified data element
- 47. Write a program to create a singly linked list of n nodes and display it in reverse order.
- 48. Sort the elements in a linked list using
 - a. changing the values (swapping the values)
 - b. Changing the address (Swapping the address)

Section 13: Polynomial using Linked List

- 49. Polynomial using linked list - addition and multiplication
- 50. Linked list using names - insert, delete, display, sort, reverse, count

Section 14- linked list

- 51. Perform the respective operations on **Linked Stack**
- 52. Perform the respective operations on **Linked Queue**
- 53. Perform the respective operations on **Circular Linked List**
- 54. Perform the respective operations on **Circular Linked Queue**

- 55. Perform the respective operations on **Doubly Linked List**
- 56. Perform the respective operations on **Circular doubly linked list – store string values as data part**
- 57. Perform the respective operations on **Circular doubly linked list – store string values as data part**

Section 15: Linear and Non Linear Search

- 58. Implement linear search for finding an element in an array.
- 59. Apply binary search for searching an element in an array of integers
- 60. Apply binary search for searching an element in an array of Strings

Section 16: Binary Search Tree

- 61. Binary search tree insertion and display Traversal using inorder, preorder and postorder using recursion
- 62. Binary search tree insertion and display in-order without using recursion
- 63. Binary search tree insertion and display pre-order without using recursion
- 64. Binary search tree insertion and display post-order without using recursion
- 65. Binary search tree insertion using names and display the names in ascending order using inorder traversal.

Section -17: Graphs

- 66. Demonstrate the data structure of adjacent matrix using arrays
- 67. Demonstrate the data structure of adjacent matrix using linked lists